

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 17

Status: active BA4 execution plan after BA4-T1 projection-boundary/traceability pressure testing.

Supersedes: Revision 16 only for forward execution state; R1-R16 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3 provenance/derivation/identity/lifecycle/change-revalidation mechanics: CLOSED BY BA3-T4.
- BAREferent and BAProposition: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. BA4 objective

BA4 defines the smallest contract by which one accepted Base Analysis can support multiple derived human and methodology-oriented views without making any view project authority or allowing consumer-specific taxonomies to redefine the shared core.

BA4 remains representation-independent and does not design a general system-modeling notation.

3. BA4-T1 - projection boundary, traceability and semantic-preservation lower-bound trial

Status: COMPLETED / PROVISIONAL PASS WITH PROJECTION-BOUNDARY LOWER-BOUND.

T1 used one human-oriented projection and one bounded flow-oriented method projection over the same accepted facial-access BA meaning, with order/provider controls.

Main dispositions

projection derived from accepted BA	REQUIRED
immutable projection revision	REQUIRED
selection/coverage contract	REQUIRED
omission == negation/absence	REJECTED
meaning-bearing projection item trace to BA	REQUIRED
mandatory duplicate governed-source provenance	REJECTED
local item identity	REQUIRED WHEN MEANING-BEARING
role-bound multi-input trace	REQUIRED WHERE ROLES DIFFER
semantic-preserving selection/rename/group	PASS
stronger shared aggregation	REJECTED
method-owned interpretation	ALLOWED DOWNSTREAM
method interpretation promoted into BA	REJECTED
projection item cross-baseline lifecycle	NOT REQUIRED BY T1
BA change -> baseline-scoped projection rebuild	REQUIRED
new BAE family	NOT FORCED
new BA2 operator	NOT FORCED

BA1 / BA2 / BA3 reopen

NOT TRIGGERED

Active provisional shape

```

BAProjectionDescriptor          [NOT BAE]
|- projectionKey
|- projectionRevisionKey        immutable
|- consumerPurpose
|- selectionCoverageContract
|- mappingContract
'- interpretationBoundary

```

```

BAProjectionMaterialization     [NOT BAE]
|- projectionRef
|- sourceBABaselineKey
'- item 0..*

```

```

BAProjectionItem               [projection-local]
|- projectionItemKey
|- projectionOwnedKind
|- traceBinding 1..*
|   |- traceRoleKey
|   '- baElementRef
|- sharedSemanticRendering 0..1
'- methodOwnedInterpretation 0..1

```

No one physical representation is mandated.

4. T1 boundary invariants to preserve under further pressure

1. a projection does not become project authority;
2. shared semantic rendering must not strengthen, invert or erase material BA meaning;
3. method-owned interpretation stays explicitly downstream;
4. omission is interpreted only under an inspectable coverage contract;
5. source drill-down remains projection -> BA -> BA3 provenance -> governed source;
6. projection-local identity is not BAE identity;
7. projection rebuild follows accepted BA change rather than creating a second project lifecycle model.

These are provisional until BA4 closure.

5. Why BA4 remains open

T1 used only one bounded method-oriented mapping family. It does not yet prove that the boundary survives when two methods impose incompatible taxonomies, when projection selection becomes highly lossy, or when method-owned interpretations are more strongly derived.

The current lower bound also needs pressure against diagnostic/stale views and cross-projection consistency after BA change.

6. BA4-T2 - method-owned interpretation, coverage loss and cross-projection consistency pressure test

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA4-T2 must perform only the next projection pressure step.

It must test at least:

1. two bounded method-oriented projections with intentionally incompatible local taxonomies derived from the same accepted BA;
2. whether each method can add its own interpretation without forcing method semantics back into BA or into a shared projection ontology;
3. whether the revisioned projection descriptor plus role-bound BA trace is sufficient to review/rebuild method-owned interpretation;
4. aggressively lossy selection where omission/completeness errors are likely;
5. a projection intentionally exposing `DIAGNOSTIC_UNRESOLVED`, `STALE` or review-oriented BA state without converting it into project truth;
6. M1-M4 plus order/WMS/provider rebuild behavior across projection families;
7. whether cross-projection comparison needs any new shared identity or relation beyond BA trace; and
8. whether a material counterexample forces only the smallest BA1/BA2/BA3/BA4 responsibility reopen.

BA4-T2 guardrails

Do not:

- define a complete STRIDE or STRIDE-AI ontology/schema;
- define AnalysisRecord or Common Finding;
- make DFD/threat-method categories BA1 types;
- introduce a universal projection ontology merely to reconcile method taxonomies;
- design ThreatForge classes/tables/UI;
- reopen BA0-BA3 without a concrete shared-semantic counterexample;
- start BA5 or BA6.

BA4-T2 exit condition

Produce a falsifiable refinement or rejection of the T1 projection lower bound and identify the next smallest BA4 pressure target. Do not close BA4 merely because two method projections can be rendered.

7. BA5 - lexical vocabulary and optional assistance

Only after BA4 boundaries are stable, define display labels, authoring synonyms and optional assistance. Preserve:

```
source lexical wording
    !=
stable BA semantic key
    !=
display label / authoring synonym
```

BA4 projection-local labels do not preempt BA5 lexical design.

8. BA6 - complete Base Analysis regression and closure

Regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout. BA6 may reopen only the smallest earlier responsibility on a material counterexample.

Only BA6 may close the complete Base Analysis milestone for the current thesis scope.

9. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

10. Next authorized microstep

Only after the BA4-T1 package is reviewed, committed, pushed and remotely verified, execute:

BA4-T2 - method-owned interpretation, coverage loss and cross-projection consistency pressure test.

Do not start BA5, BA6 or downstream analysis schemas before BA4-T2 is completed and reviewed.